Python
INTERVIEWER-STYLE PRACTICE

Fresher Data Analyst | English answers + Hindi explanations + practical examples + follow-up cross questions

20 interview scenarios | Practice answering aloud | Use only project details you have actually verified

How to practice

Read the question first and answer aloud before reading the model answer. Learn the concept, not just the wording. For syntax, mention that exact functions may vary by tool or database version. For project questions, describe only work you personally completed.

Interviewer Q1: Why is Python used in data analytics?

Suggested English answer: Python has readable syntax and libraries for cleaning, analysis, visualization, automation, and connecting to data sources.

Hindi me samjho: Python se data clean, analyze, visualize aur repetitive tasks automate kar sakte hain.

Interviewer follow-up: Which libraries have you used?

How to answer: Name only those you have practiced, such as pandas, NumPy, Matplotlib, or Seaborn.

Interviewer Q2: Explain list, tuple, set, and dictionary.

Suggested English answer: A list is ordered and mutable; tuple is ordered and immutable; set stores unique values; dictionary maps keys to values.

Hindi me samjho: List changeable, tuple immutable, set unique values, dictionary key-value.

Interviewer follow-up: Which would you use for employee salary lookup by ID?

How to answer: A dictionary can map employee ID to salary for quick key-based lookup.

Interviewer Q3: What is the difference between == and is?

Suggested English answer: == checks value equality; is checks object identity, meaning whether both references point to the same object.

Hindi me samjho: == value compare; is same object reference check karta hai.

Interviewer follow-up: Should you use is to compare strings?

How to answer: Usually use == for value comparison; use is for identity checks, commonly with None.

Interviewer Q4: What is a pandas DataFrame?

Suggested English answer: A DataFrame is a two-dimensional labeled structure with rows and columns, suitable for tabular data.

Hindi me samjho: Rows-columns wali table jaisa structure hai.

```
Practical example:
df = pd.read_csv('employees.csv')
```

Interviewer follow-up: How do you inspect it?

How to answer: Use head(), shape, info(), describe(), and isna().sum().

Interviewer Q5: Explain loc versus iloc.

Suggested English answer: loc selects by labels and Boolean conditions; iloc selects by integer positions.

Hindi me samjho: loc labels/condition se; iloc numeric position se.

```
Practical example:
df.loc[df['salary']>3000, ['employee_name','salary']]
```

Interviewer follow-up: How would you select first five rows and first two columns with iloc?

How to answer: df.iloc[:5, :2]

Interviewer Q6: How do you filter employees earning over 3000 and show name and salary only?

Suggested English answer: I use a Boolean mask with loc and specify the required columns.

Hindi me samjho: Salary condition filter karke sirf name aur salary columns nikaalunga.

```
Practical example:
df.loc[df['salary'] > 3000, ['employee_name', 'salary']]
```

Interviewer follow-up: What does the comma separate?

How to answer: Rows selection is before the comma; columns selection is after it.

Interviewer Q7: How do you handle missing values?

Suggested English answer: I inspect missingness first, understand its cause, then drop or impute using a business-appropriate method and validate the impact.

Hindi me samjho: Pehle missing ka reason samjho, phir justified fill/drop aur validation.

```
Practical example:
df.isna().sum()
```

Interviewer follow-up: Should every missing value be replaced with zero?

How to answer: No. Zero can change meaning; use a justified strategy such as median, category, or retaining missingness.

Interviewer Q8: How do you remove duplicate rows?

Suggested English answer: Use duplicated to inspect and drop_duplicates to remove, choosing the subset and keep rule deliberately.

Hindi me samjho: Duplicate identify karke subset/keep rule decide karunga.

```
Practical example:
df.drop_duplicates(subset=['employee_id'], keep='last')
```

Interviewer follow-up: Why choose subset?

How to answer: A repeated employee ID may define a duplicate even if timestamps differ; business rules decide.

Interviewer Q9: Explain groupby with a business example.

Suggested English answer: groupby creates groups and applies aggregation, such as average salary by department or total sales by region.

Hindi me samjho: Department-wise average salary ya region-wise sales total nikalne ke liye.

```
Practical example:
df.groupby('department')['salary'].mean()
```

Interviewer follow-up: How do you get multiple metrics?

How to answer: Use agg, for example sum, mean, and count on selected columns.

Interviewer Q10: What is the difference between merge and concat?

Suggested English answer: merge combines tables by matching keys; concat stacks rows or columns along an axis.

Hindi me samjho: merge key se join; concat rows/columns me append karta hai.

Interviewer follow-up: Which is like SQL JOIN?

How to answer: pandas.merge is the closest general equivalent.

Interviewer Q11: How do you sort salaries descending?

Suggested English answer: Use sort_values with ascending=False.

Hindi me samjho: Salary high-to-low sort karne ke liye.

```
Practical example:
df.sort_values('salary', ascending=False)
```

Interviewer follow-up: How do you reset the index afterward?

How to answer: Use reset_index(drop=True) if the old index is not needed.

Interviewer Q12: What is NumPy used for?

Suggested English answer: NumPy provides efficient arrays and vectorized numerical operations.

Hindi me samjho: Fast numerical array aur calculations ke liye.

Interviewer follow-up: Why can vectorization help?

How to answer: It reduces Python-level loops and uses optimized array operations.

Interviewer Q13: What is a lambda function?

Suggested English answer: A lambda is a small anonymous function for a short expression, often used in transformations.

Hindi me samjho: Short one-expression function.

```
Practical example:
df['name_len'] = df['name'].apply(lambda x: len(str(x)))
```

Interviewer follow-up: Would you use lambda for complex logic?

How to answer: A named function is clearer for complex or reusable logic.

Interviewer Q14: How do you read and export CSV?

Suggested English answer: Use read_csv to load and to_csv to export; index=False avoids exporting the DataFrame index as an extra column.

Hindi me samjho: CSV load/export pandas se; index=False extra index column rokta hai.

```
Practical example:
df = pd.read_csv('input.csv')
df.to_csv('output.csv', index=False)
```

Interviewer follow-up: How do you read Excel?

How to answer: Use pd.read_excel with the correct sheet_name and installed engine.

Interviewer Q15: What is exception handling?

Suggested English answer: try/except catches expected errors and lets the program respond safely; avoid hiding errors broadly without logging.

Hindi me samjho: Expected error ko handle karo, lekin silently hide mat karo.

Interviewer follow-up: Why not use a bare except?

How to answer: It can hide programming errors and make debugging difficult.

Interviewer Q16: What is a function?

Suggested English answer: A function is reusable code defined with def; it can accept parameters and return a value.

Hindi me samjho: Reusable code block.

```
Practical example:
def add_tax(amount, rate):
return amount * (1 + rate)
```

Interviewer follow-up: What is the difference between print and return?

How to answer: print displays output; return sends a value back to the caller.

Interviewer Q17: How would you clean a sales dataset?

Suggested English answer: Inspect columns and types, standardize names/categories, parse dates, handle missing values and duplicates, validate totals, then save a reproducible output.

Hindi me samjho: Schema/type check, labels standardize, date parse, null/duplicate handle, totals validate.

Interviewer follow-up: How do you prove cleaning did not corrupt data?

How to answer: Compare row counts, key uniqueness, totals, and sample records before and after.

Interviewer Q18: What is a Jupyter Notebook?

Suggested English answer: It is an interactive environment for running code cells alongside outputs, notes, and charts.

Hindi me samjho: Code aur output cells me run karne ka environment.

Interviewer follow-up: What should a clean notebook include?

How to answer: Clear steps, concise markdown, reproducible inputs, and outputs that can be rerun.

Interviewer Q19: How do you connect Python to SQL?

Suggested English answer: Use a supported database driver or SQLAlchemy, parameterize queries, protect credentials, and close connections reliably.

Hindi me samjho: Driver/SQLAlchemy se connect; credentials safe aur parameterized query.

Interviewer follow-up: Why parameterize?

How to answer: It helps prevent SQL injection and handles values safely.

Interviewer Q20: How would you explain your Python project in an interview?

Suggested English answer: I would state the question, data source, cleaning steps, analysis performed, key verified finding, and limitation.

Hindi me samjho: Problem, dataset, cleaning, analysis, verified result aur limitation explain karunga.

Interviewer follow-up: What if they ask for a result you cannot remember?

How to answer: Be honest and explain how you would reproduce or verify it.

Final interview reminder: Pause, clarify the requirement, explain your approach, and validate the result. It is acceptable to say you would verify a detail rather than guess.